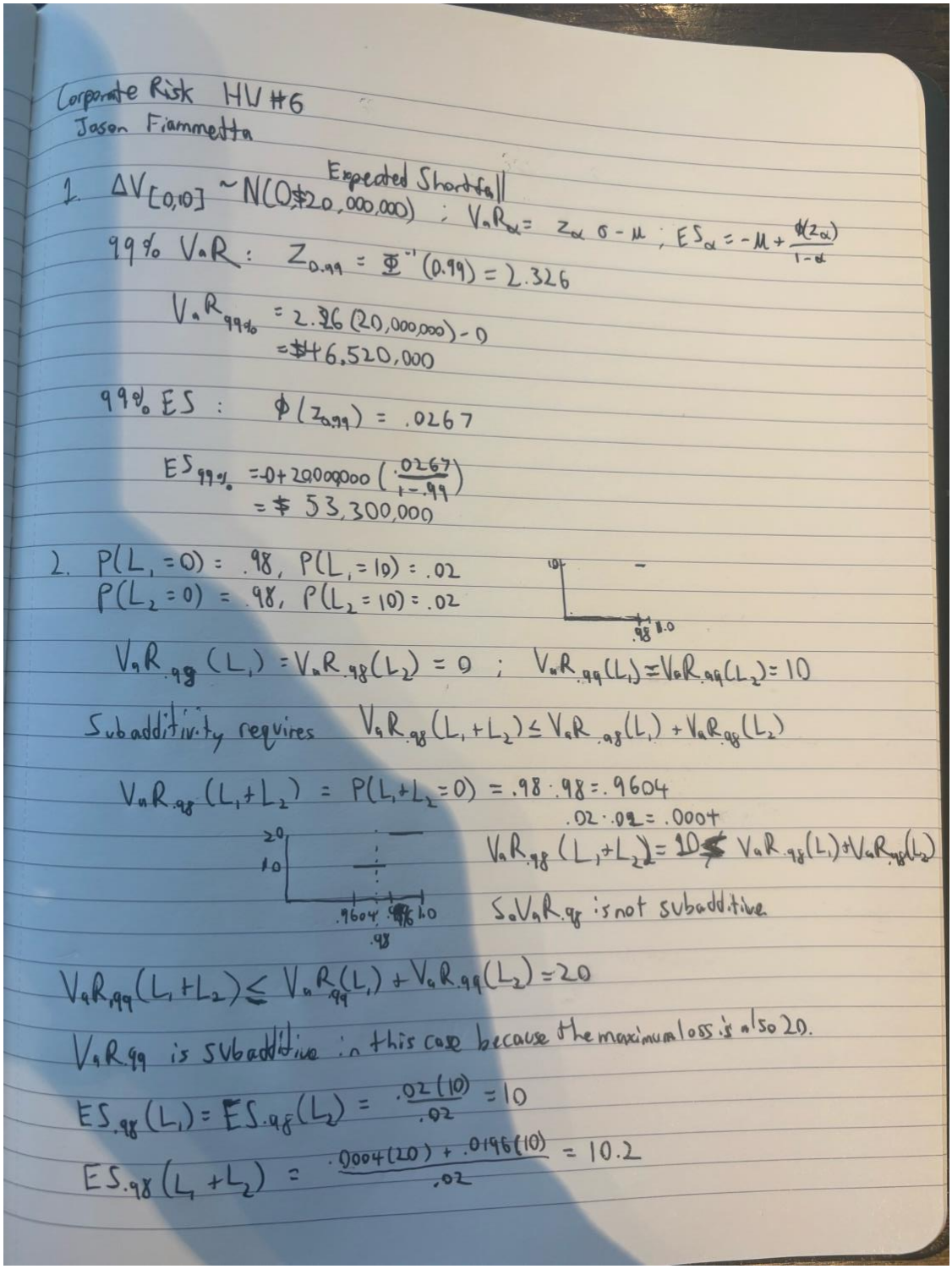
Corporate Risk Management

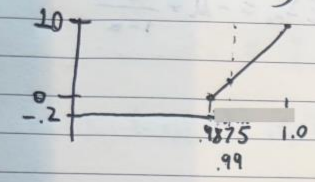
Homework #4 Due October 13th, 2026

Jason Fiammetta



3. Two $10 million 1-year loans. $P(\text{loan}_1 \text{ defaults}) = P(\text{loan}_2 \text{ defaults}) = 1.25\%$
$P(\text{neither defaults}) = 97.5\%$ $P(\text{both default}) = 0\%$

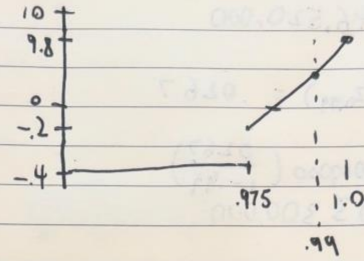
Non-defaulting loan profits $200,000



$$VaR_{.99}(L_1) = VaR_{.99}(L_2) = \frac{\$10{,}000{,}000\,(.99 - .9875)}{(1.0 - .9875)}$$

$$= \$2{,}000{,}000$$

L_1 / L_2	no default	default
no default	97.5	1.25
default	1.25	0

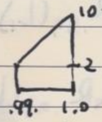


$$VaR_{.99}(L_1 + L_2) = \frac{\$10{,}000{,}000\,(.99 - .975)}{(1.0 - .975)} - \$200{,}000$$

$$= \$5{,}800{,}000 \quad > VaR_{.99}(L_1) + VaR_{.99}(L_2)$$

$\Rightarrow VaR_{.99}(L_1 + L_2)$ is not subadditive

$$ES_{.99}(L_1) = ES_{.99}(L_2) =$$



$$= \frac{\$2{,}000{,}000 \cdot .01 + \frac{1}{2} \cdot .01 \cdot \$8{,}000{,}000}{.01} = \$6{,}000{,}000$$

$$ES_{.99}(L_1 + L_2) =$$



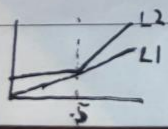
$$= \frac{\$5{,}800{,}000 \cdot .01 + \$4{,}900{,}000 \cdot \frac{1}{2} \cdot .01}{.01} = \$7{,}800{,}000$$

4. VaR monotinicity says $L_1 \leq L_2 \Rightarrow VaR_\alpha(L_1) \leq VaR_\alpha(L_2)$.

This does not imply $L_1 < L_2 \Rightarrow VaR_\alpha(L_1) < VaR_\alpha(L_2)$

Counter example: $U \sim Unif(0,1)$; $L_1 = U$; $L_2 = U + |U - \frac{1}{2}|$

$L_1 < L_2$ P.a.s; $P(U = \frac{1}{2}) = 0$; $VaR_{.5}(L_1) = VaR_{.5} = .5$



5. $VaR_\alpha(L)$ and $ES_\alpha(L)$ for:

a) $L \sim e^{1/\theta}$

$VaR_\alpha(L) =$ first x such that $F_L(x) \geq \alpha$

$f_L(x) = \theta e^{-\frac{x}{\theta}} \Rightarrow F_L(x) = 1 - e^{-\frac{x}{\theta}} \equiv \alpha$

$\Rightarrow e^{-\frac{x}{\theta}} = 1-\alpha$

$x = -\theta \ln(1-\alpha) = VaR_\alpha(L)$

$ES_\alpha(L) = \mathbb{E}[L \mid L > VaR_\alpha(L)]$

$= \mathbb{E}[L] + VaR_\alpha(L) = \theta - \theta \ln(1-\alpha)$

b) $L \sim$ Pareto $(1, 1/\xi)$ with CDF $F(x) = \begin{cases} 0 & x \leq 1 \\ 1 - x^{-\frac{1}{\xi}} & x > 1 \end{cases}$

How does the answer depend on ξ?

$f_L(x) = \begin{cases} 0 & x \leq 1 \\ \frac{1}{\xi} x^{-\frac{1}{\xi}-1} & x > 1 \end{cases}$ $\quad VaR_\alpha(L)$ $v \Rightarrow F_L(v) = \alpha$

$1 - v^{-\frac{1}{\xi}} = \alpha$

$v = (1-\alpha)^{-\xi} = VaR_\alpha(L)$

$ES_\alpha(L) = \mathbb{E}[L \mid L > VaR_\alpha(L)]$

$= \frac{1}{1-\alpha} \int_v^\infty x f_L(x) dx = \frac{1}{1-\alpha} \int_v^\infty x \frac{1}{\xi} x^{-\frac{1}{\xi}-1} dx$

The integral is only finite if $-\frac{1}{\xi} < 1 \Rightarrow 1 > \xi$

for $0 < \xi < 1$: $ES_\alpha(L) = \frac{1}{1-\alpha}\left(\frac{-v^{1-\frac{1}{\xi}}}{\frac{1}{\xi}-1}\right) = \frac{1}{1-\alpha}\left(\frac{1-\xi}{\xi^{-1}}(1-\alpha)^{-\xi+1}\right)$

$= \frac{\xi^{-1}(1-\alpha)^{-\xi}}{1-\xi}$

or if $\xi \geq 1$, $ES = \infty$